

LAB REPORT

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Date :22 /9 /2023

Hypothesis

- if an acid is added to base with phenolphthalein, the pH of the solution will dramatically decrease until it reaches pH – 7.
- If proper experiment is executed, salts and water will be formed.
- If phenolphthalein is added to the base, the color of the base will be changed.
- The color of litmus paper will not be changed if predicted product is outputted.

Materials

- Hydrochloric acid and Sodium hydroxide base.
- Conical flask and Tall beaker.
- Burette with stopcock.
- Stand with Clamper.
- Phenolphthalein, litmus blue and red.

Observation and Analysis

What colour appears when the phenolphthalein is added to the solution in the conical flask?

- Pink or dark pink.

Why does the colour disappear when the acid is added?

- The color disappears when an acid is added to a base due to the neutralisation effect of acid and the subsequent loss of H⁺ ions.

Does the solution obtained from the experiment affect the colour of either blue and red litmus paper?

- No, after the experiment salt and water will be formed as a result it will not affect the color of both blue and red litmus papers.

Write the balanced chemical equation for the reaction that takes place in this experiment.

- $\text{HCl} + \text{NaOH} \rightarrow \text{NaCl} + \text{H}_2\text{O}$.

Conclusion

- I concluded that an acid neutralise base with phenolphthalein indicator as a result salt and water are formed. consequently the color of both blue and red litmus paper is not changed.